

Computer Science & IT

Database Management System



Query Languages

Lecture No. 10



By- Vishal Sir



Recap of Previous Lecture

- ✓ **Topic** Execution w.r.t. correlated subquery
- ✓ **Topic** AS clause
- ✓ **Topic** WITH clause



Topics to be Covered



Topic

Comparison with NULL



Topic

Comparison with Regular expression



Topic

Practice questions



H.W.
#Q.



Consider the relational database with the following four schemes and their respective instances.

Student(sNo, sName, dNo) Dept(dNo, dName)

Course(cNo, cName, dNo) Register(sNo, cNo)

Student		
sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Dept	
dNo	dName
D01	CSE
D02	EEE

Course		
cNo	cName	dNo
C11	DS	D01
C12	OS	D01
C21	DE	D02
C22	PT	D02
C23	CV	D03

Register	
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

Question Continues in Next Slide

#Q.

SQL Query:

SELECT * FROM Student AS S
WHERE NOT EXIST

(SELECT cNo FROM Course WHERE dNo = "D01".
EXCEPT $\equiv$ Minus

SELECT cNo FROM Register WHERE sNo = S.sNo)

independent

{C11, C12}

The number of rows returned by the above SQL query is _____.

Correlated

H.W.
#Q.

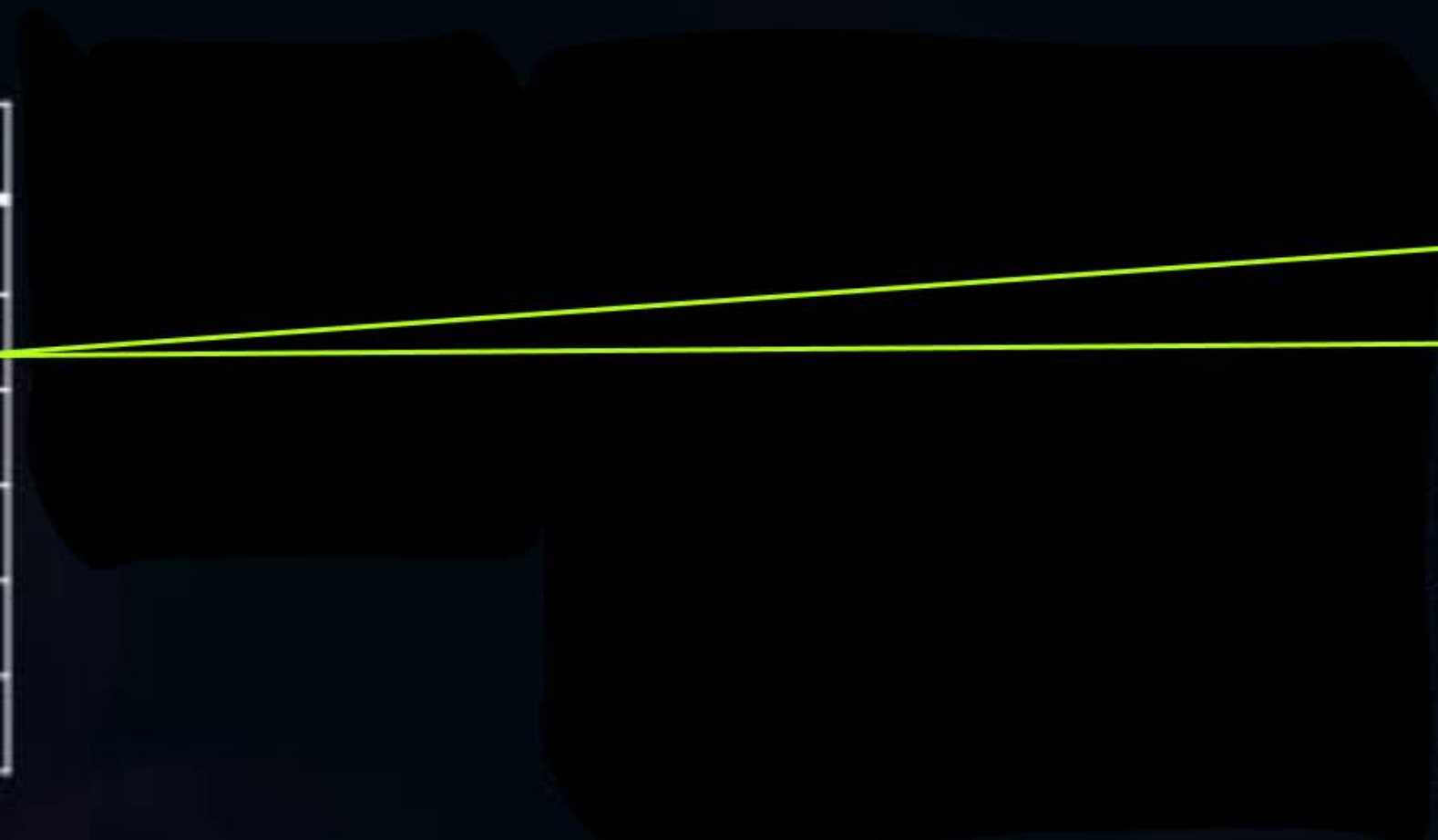


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S05	Milli	D02



Register	
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

✓
✓
x
x
x
x
x
x
x
x

Question Continues in Next Slide

H.W.
#Q.



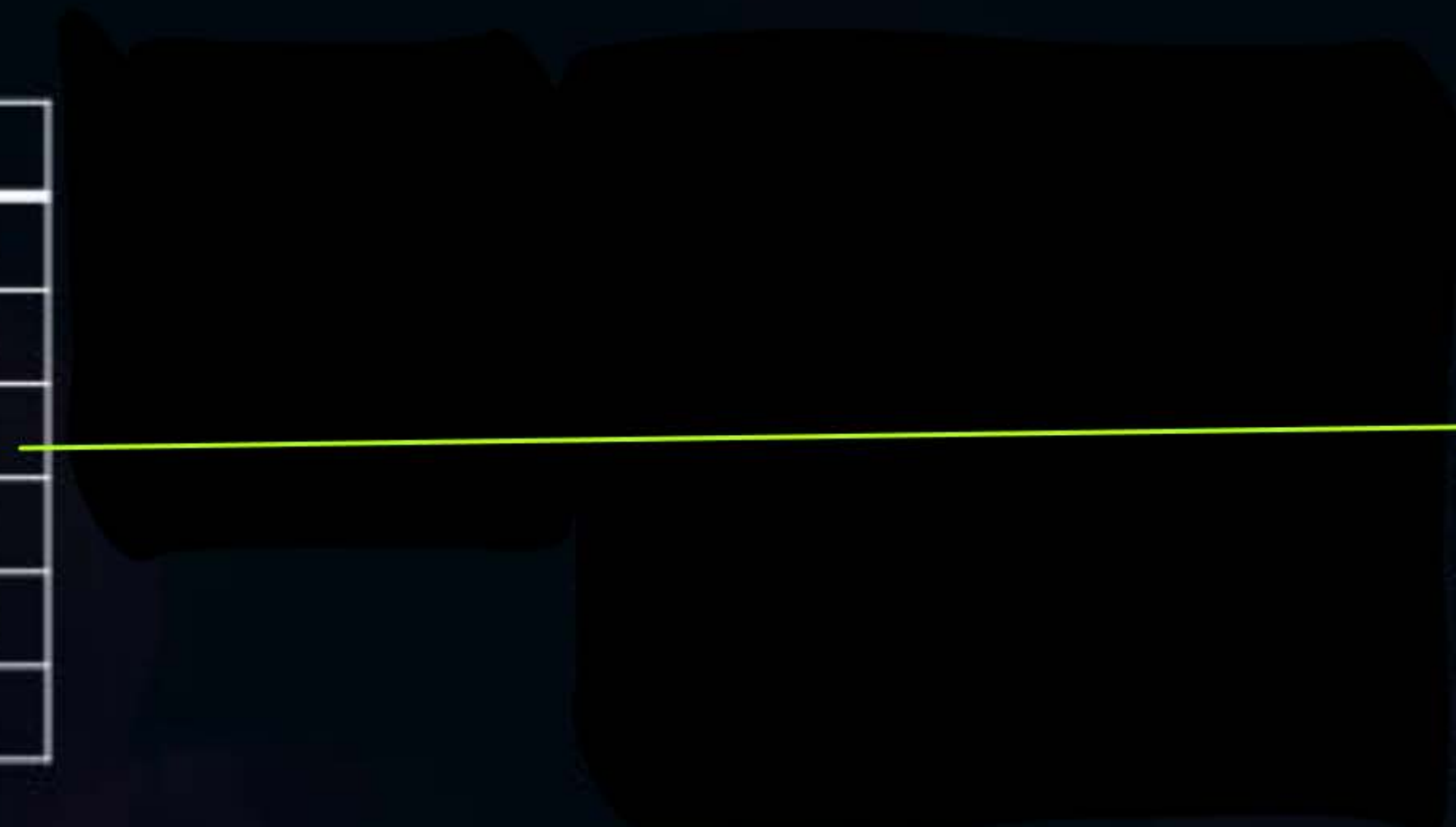
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S04	Jane	D01
S05	Milli	D02

Register	
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
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Question Continues in Next Slide

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S02	C11
S03	C21
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Question Continues in Next Slide

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S05	Milli	D02

Register	
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

Question Continues in Next Slide

#Q. SQL Query:

SELECT * FROM Student AS S
WHERE NOT EXIST

(SELECT cNo FROM Course WHERE dNo = "D01".

EXCEPT $\equiv$ Minus

(SELECT cNo FROM Register WHERE sNo = S.sNo)

The number of rows returned by the above SQL query is 2.

Two tuples in o/p

S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Mili	D02

{C ₁₁ , C ₁₂ }	-	{C ₁₁ , C ₁₂ }	=	$\emptyset$
{C ₁₁ , C ₁₂ }	-	{C ₁₁ }	=	{C ₁₂ }
{C ₁₁ , C ₁₂ }	-	{C ₂₁ , C ₂₂ , C ₂₃ }	=	{C ₁₁ , C ₁₂ }
{C ₁₁ , C ₁₂ }	-	{C ₁₁ , C ₁₂ }	=	$\emptyset$
{C ₁₁ , C ₁₂ }	-	{C ₁₁ , C ₂₁ }	=	{C ₁₂ }

Result of NOT EXISTS
True
False
False
True
False



Topic : Comparison with NULL

- NULL is a non zero Unknown Value,
and no two NULL are equal

To Compare with "NULL"
operator used is "IS"

Complement of "IS" is "IS NOT"

EMP

Eid	Ename	Pass-NO.
S ₁	A	123
S ₂	B	438
S ₃	A	NULL
S ₄	C	275

Retrieve Eids of the Employee with no passport.

→ Select Eid
From EMP
Where Pass-NO. = 0

Wrong because
NULL is "not
equal to "Zero"

→ Select Eid
From EMP
Where Pass-NO. = NULL

Wrong because
no two NULL
are equal.

Correct query:-

Select Eid
From EMP
Where Pass-NO. IS NULL.

EMP

Eid	Ename	Pass-NO.
S ₁	A	123
S ₂	B	438
S ₃	A	NULL
S ₄	C	275

Retrieve Eids of the
Employee with valid passport No.

Select Eid
From EMP

Where Pass-No. IS NOT NULL



Topic : Regular expression

' _ ' $\Rightarrow$ Underscore represent exactly one alphabet

' % ' $\Rightarrow$ % represent '0' or more alphabets
i.e., any no. of

* Write the regular Expression for the names starting with "R".

Ans: ' R % '

first alphabet must be R

after that there may be any no. of alphabets and they may be anything



Topic : Comparison with regular expression

- To Compare with regular Expression
operator used is "LIKE"

Complement of "LIKE" is "NOT LIKE"

EMP

Eid	Ename	Pass-NO.
S ₁	Ram	123
S ₂	Radha	438
S ₃	Ramesh	NULL
S ₄	Roman	275
S ₅	Sumit	376

Retrieve Eids of all the Employee
Whose names start with R

Select Eid
from EMP
Where Ename LIKE 'R,%'

EMP

	Eid	Ename	Pass-NO.
✓	S ₁	Ram	123
✗	S ₂	Radha	438
✓	S ₃	Ramesh	NULL
✓	S ₄	Roman	275
✗	S ₅	Sumit	376

Retrieve Eids of all the Employee
Whose names start with R
and third alphabet in the name is 'm'

Select Eid
from EMP

Where Ename LIKE 'R,_,m,%'

EMP

	Eid	Ename	Pass-NO.
✗	S ₁	Ram	123
✗	S ₂	Radha	438
✓	S ₃	Ramesh	NULL
✓	S ₄	Roman	275
✗	S ₅	Sumit	376

Retrieve Eids of all the Employee

Whose names start with R
and third alphabet in the name is 'm'
and the name must contain at least 5 alphabets.

Select Eid
from EMP

Where Ename LIKE 'R,_,m,_,_,%'

#Q. Consider The Following Relational Scheme

- ✓ Student (school-id, sch-roll-no, sname, saddress)
- ✓ School (school-id, sch-name, sch-address, sch-phone)
- ✓ Enrolment (school-id, sch-roll-no, erollno, examname)
- ✓ ExamResult (Errollno, examname, marks)

(A) for each school with more than 200 students appearing in exams, the name of the school and the number of 100s scored by its students

(B) for each school with more than 200 students in it, the name of the school and the number of 100s scored by its students

(C) for each school with more than 200 students in it, the name of the school and the number of its students scoring 100 in at least one exam

✓ (D) nothing; the query has a syntax error

H.W.

#Q.

What does the following SQL query output?

```
SELECT sch-name, COUNT (*)  
FROM School C, Enrolment E, ExamResult R  
WHERE E.school-id = C.school-id  
      AND  
      E.examname = R.examname  
      AND  
      E.erollno = R.erollno  
      AND  
      R.marks = 100  
      AND  
      E.school-id IN  
      (SELECT school-id  
       FROM student GROUP BY school-id HAVING COUNT (*) > 200)  
GROUP BY school-id
```

Not in Group by

#Q.

Consider the following relational schema:

[MCQ: 2009: 2M]



- Suppliers(sid:integer, sname:string, city:string, street:string)
- Parts(pid:integer, pname:string, color:string)
- Catalog(sid:integer, pid:integer, cost:real)

Consider the following relational query on the above database:

```
SELECT S.sname
FROM Suppliers S
WHERE S.sid NOT IN
  (SELECT C.sid
   FROM Catalog C
   WHERE C.pid NOT IN
     (SELECT P.pid
      FROM Parts P
      WHERE P.color <> 'blue'))
```

if supplier has supplied any blue part then sid will get selected

o/p will be
Pids of all non-blue color parts
o/p =

P₁
P₃

Supplier

Sid
S ₁
S ₂
S ₃

Catalog

Sid	Pid
S ₁	P ₁
S ₁	P ₂
S ₂	P ₃

Parts

Pid	Color
P ₁	Red
P ₂	Blue
P ₃	Red

Question Continues in Next Slide

#Q. Assume that relations corresponding to the above schema are not empty. Which one of the following is the correct interpretation of the above query?

- A. $\frac{\%}{p} \cdot \begin{bmatrix} S_1 \\ S_2 \end{bmatrix}$ Find the names of all suppliers who have supplied a non-blue part.
- B. $\begin{bmatrix} S_3 \end{bmatrix}$ Find the names of all suppliers who have not supplied a non-blue part.
- C. $\begin{bmatrix} \emptyset \end{bmatrix}$ Find the names of all suppliers who have supplied only blue parts.
- D. $\begin{bmatrix} S_1 \end{bmatrix}$ Find the names of all suppliers who have not supplied only blue parts.

Correct ans will be:

Find the names of all the suppliers who did not supply any blue color part.

i.e. they have supplied parts of other colors as well along with some blue color parts



2 mins Summary



✓ **Topic**

Comparison with NULL

✓ **Topic**

Comparison with Regular expression

✓ **Topic**

Practice questions

THANK - YOU